

STAGE 4 TECHNOLOGY MANDATORY

Catapult | 25-period teaching program

Engineering technologies and systems | Syllabus-aligned network review copy v2.0

Course	Stage 4 Technology Mandatory	Syllabus	Technology 7-8 Syllabus (2023)
Delivery	10 weeks 25 x 60-minute periods	Pattern	5 periods per fortnight
Lesson balance	10 Site/theory 5 Project/folio 10 Practical	Implementation	Current and mandatory from 2026
Official source	NSW Curriculum official page verified 28 Aug 2026	Focus	Engineering technologies and systems
Status	Local review copy matching Word, PDF and manifest v2.0	Approval	Teacher to confirm before delivery
Complete Stage 4 outcomes appear below. Period rows identify the outcomes directly supported by each lesson; this does not claim that every outcome is formally assessed in this unit.			

Official Stage 4 outcomes

Outcome	Exact official wording	Outcome	Exact official wording
TE4-SDP-01	explains relationships between sustainability, design and production	TE4-DES-01	communicates and evaluates design ideas and solutions
TE4-PDP-01	describes the practices and processes of designers and producers	TE4-SAF-01	selects and safely uses tools, materials, technologies and processes
TE4-MSG-01	explains how materials, systems and components contribute to solutions	TE4-DIG-01	demonstrates technological literacy to safely interact in digital environments
TE4-PPM-01	applies processes in the planning, management and production of projects	TE4-DIG-02	uses data and digital systems to code, design and produce projects

What students are learning

Learning focus	In plain English	Learning focus	In plain English
Sustainability, design and production	Explain how design and production choices affect sustainability.	Design communication and evaluation	Communicate ideas and judge solutions against criteria.
Designers and producers	Describe how people investigate, plan, make and improve solutions.	Safe practical work	Choose and use approved tools, materials, technologies and processes safely.
Materials, systems and components	Explain how parts, materials and systems affect how a solution works.	Safe digital literacy	Work safely and responsibly in digital environments.
Project planning and production	Plan, manage and produce the project in a clear sequence.	Data, coding and digital systems	Use data or digital systems where they support the unit's design and production.

Teacher-to-confirm register

When	Decision needed	Confirm by
Before the unit	Timetable, lesson order, class supports and approved adjustments	Teacher / faculty
Before practical work	Activity, materials, equipment, supervision and current risk controls	Teacher / faculty
Before assessment	Task authority, marks, dates, conditions and submission method	Teacher / faculty
Before classroom issue	Current links, program version and any local sequence changes	Teacher / faculty

Teaching program | Periods 1-10

10 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

P	Type	Focus	Teaching and learning	Syllabus alignment	What the teacher checks
1	<u>Site/theory</u>	Catapult as an engineered system	Teach A system is more than a set of parts; Inputs, processes, outputs and feedback.	component properties engineered products and systems Outcomes: TE4-MSC-01, TE4-PDP-01	4 section checks, bank-01
2	<u>Project/folio</u>	Brief, constraints and evidence	Reading the brief as an engineer.	design factors design and systems thinking Outcomes: TE4-DES-01, TE4-PPM-01	2 section checks, ala-m01, m01-capstone, folio-01, bank-02
3	Practical	Workshop orientation and readiness	Teacher-led: Workshop orientation and readiness. Local procedure, equipment and controls TBC.	safe practices Outcomes: TE4-SAF-01, TE4-PPM-01	teacher observation, folio-05
4	<u>Site/theory</u>	Tools, materials and planned safety	Teach Safety is a planned system; Select tools by function; Pine, grain and elastic behaviour.	materials/processes and skills materials/components/systems Outcomes: TE4-SAF-01, TE4-PDP-01	6 section checks, ala-m02, m02-capstone, folio-02, bank-03
5	Practical	Teacher-approved material and tool-function practice	Teacher-led: Teacher-approved material and tool-function practice. Local procedure, equipment and controls TBC.	select components and systems safe practices Outcomes: TE4-SAF-01, TE4-PPM-01	teacher observation, folio-05
6	<u>Site/theory</u>	Energy stores and lever moments	Teach Different systems, different energy stores; Lever moments and trade-offs.	force, motion and energy engineered products and systems Outcomes: TE4-MSC-01, TE4-PDP-01	4 section checks, bank-04
7	<u>Project/folio</u>	Woomera context and design reasoning	A woomera in its own cultural and technological context. Named sources only.	First Nations engineered systems design and systems thinking Outcomes: TE4-DES-01, TE4-PPM-01	2 section checks, ala-m03, m03-capstone, folio-02, bank-12
8	Practical	Teacher-controlled lever observation	Teacher-led: Teacher-controlled lever observation. Local procedure, equipment and controls TBC.	force, motion and energy Outcomes: TE4-SAF-01, TE4-PPM-01	teacher observation, folio-02
9	<u>Site/theory</u>	Force, energy, Newton's laws and fair prototypes	Teach Following the energy chain; Newton's three laws in one system; Prototypes and fair evidence.	force, motion and energy prototype and working-model evaluation Outcomes: TE4-MSC-01, TE4-PDP-01	6 section checks, bank-05
10	Practical	Prototype question and fair evidence	Teacher-led: Prototype question and fair evidence. Local procedure, equipment and controls TBC.	prototype and working-model evaluation test, modify and improve Outcomes: TE4-SAF-01, TE4-PPM-01	teacher observation, ala-m04, m04-capstone, folio-07
Teacher to confirm before use: Periods 1-10 practical activities, materials, equipment, supervision and current risk controls.					

Teaching program | Periods 11-20

10 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

P	Type	Focus	Teaching and learning	Syllabus alignment	What the teacher checks
11	<u>Site/theory</u>	Ideas and drawings that agree	Teach Develop, compare and select ideas; Drawings that agree with each other.	design and systems thinking graphical communication Outcomes: TE4-DES-01, TE4-PDP-01	4 section checks, folio-03
12	<u>Project/folio</u>	Cutting list, material arithmetic and design approval	Cutting lists and material limits.	data and information document design/production Outcomes: TE4-DES-01, TE4-PPM-01	2 section checks, ala-m05, m05-capstone, folio-04, bank-06
13	Practical	Approved transition from design to making	Teacher-led: Approved transition from design to making. Local procedure, equipment and controls TBC.	materials/processes and skills select components and systems Outcomes: TE4-SAF-01, TE4-PPM-01	teacher observation, folio-04, folio-06
14	<u>Site/theory</u>	Production sequence, load paths and quality evidence	Teach Sequence work around dependencies; Structures carry forces through load paths; Check quality and record real evidence.	document design/production production quality Outcomes: TE4-MSC-01, TE4-PDP-01	6 section checks, ala-m06, m06-capstone, folio-05, bank-07
15	Practical	Base/frame production and checkpoint evidence	Teacher-led: Base/frame production and checkpoint evidence. Local procedure, equipment and controls TBC.	materials/processes and skills safe practices Outcomes: TE4-SAF-01, TE4-PPM-01	teacher observation, folio-06
16	<u>Site/theory</u>	Component interaction, alignment and quality	Teach Components form an interaction chain; Quality depends on alignment and fit; Stop, check, adjust and record.	select components and systems production quality Outcomes: TE4-MSC-01, TE4-PDP-01	6 section checks, ala-m07, m07-capstone
17	Practical	Teacher-approved assembly and quality checks	Teacher-led: Teacher-approved assembly and quality checks. Local procedure, equipment and controls TBC.	materials/processes and skills safe practices Outcomes: TE4-SAF-01, TE4-PPM-01	teacher observation, folio-06
18	<u>Project/folio</u>	Production log and quality evidence conference	Check quality and record real evidence; Stop, check, adjust and record.	document design/production justify tested materials/components Outcomes: TE4-DES-01, TE4-PPM-01	4 section checks, folio-05, folio-06, bank-08
19	<u>Site/theory</u>	Testable questions, repeated data and redesign	Teach Begin with a testable question; Use repeated data, mean and range; Reason from evidence and redesign carefully.	prototype and working-model evaluation testing in an engineering report Outcomes: TE4-DES-01, TE4-PDP-01	6 section checks, bank-09
20	Practical	Teacher-controlled testing and one redesign decision	Teacher-led: Teacher-controlled testing and one redesign decision. Local procedure, equipment and controls TBC.	justify tested materials/components test, modify and improve Outcomes: TE4-SAF-01, TE4-PPM-01	teacher observation, ala-m08, m08-capstone, folio-07

Teacher to confirm before use: Periods 11-20 practical activities, materials, equipment, supervision and current risk controls.

Teaching program | Periods 21-25

5 x 60-minute lessons. Text is 9.5 pt and wraps within each cell.

P	Type	Focus	Teaching and learning	Syllabus alignment	What the teacher checks
21	Site/theory	Engineering report and evidence-based judgement	Teach Build a report around the design journey; Use claim, evidence and reasoning.	testing in an engineering report design factors and quality Outcomes: TE4-DES-01, TE4-PDP-01	4 section checks, bank-10
22	Practical	Final teacher-approved adjustment and quality evidence	Teacher-led: Final teacher-approved adjustment and quality evidence. Local procedure, equipment and controls TBC.	production quality design factors and quality Outcomes: TE4-SAF-01, TE4-PPM-01	teacher observation, folio-06, folio-08
23	Project/folio	Sustainability, report completion and assessment preparation	Think across environmental, social and economic needs.	sustainability evaluation society and sustainability Outcomes: TE4-DES-01, TE4-PPM-01	2 section checks, ala-m09, m09-capstone, folio-08, bank-11
24	Site/theory	Demonstration, evaluation and connected systems	Teach Demonstrate function through evidence; Evaluate product, process and learning; Connected Aboriginal engineering systems. Named sources only.	First Nations engineered systems society and sustainability Outcomes: TE4-MSC-01, TE4-PDP-01	6 section checks, ala-m10, m10-capstone, folio-08, bank-12
25	Practical	Teacher-controlled demonstration and final evidence check	Teacher-led: Teacher-controlled demonstration and final evidence check. Local procedure, equipment and controls TBC.	design factors and quality test, modify and improve Outcomes: TE4-SAF-01, TE4-PPM-01	teacher observation, folio-08
Teacher to confirm before use: Periods 21-25 practical activities, materials, equipment, supervision and current risk controls.					

Evidence, feedback and support

Evidence	Look for	Respond / support
Theory and module work	Understanding and use of key ideas	Return to the named section and model one stronger response
Project and folio	Design decisions, planning and refinement	Conference against criteria and give one clear next step
Practical observation	Safety, workflow, skill, quality and clean-down	Pause, demonstrate and recheck before continuing
Product and evaluation	Quality and evidence-based judgement	Compare the result with the original criteria